

# Enabling Consumer UAVs for Precision Agriculture Applications: A Case Study of Yield Estimation

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**Abstract**—Unmanned aerial vehicles (UAVs) equipped with various sensors and onboard processing capabilities have emerged as a promising means to acquire field data for precision agriculture applications. However, such UAVs are costly, restricting their deployment in small-to-medium-sized fields, particularly in developing countries. In contrast, consumer-grade UAVs have high-resolution RGB cameras and video streaming abilities at affordable prices. This paper presents an efficient processing pipeline to analyze video streams from consumer-grade UAVs on smartphones. The processing pipeline consists of preprocessing, object detection, and yield estimation. The object detector, being the most computationally expensive module, is invoked every  $n$ th frame due to video redundancy and the target platform's limited resources. The yield estimation task on a smartphone requires efficient and accurate fruit detection, which a modified YOLOv8n model achieved. We evaluate our pipeline on datasets of apple and peach trees and demonstrate that it can process UAV-captured images to collect yield-related statistics. We also discuss the lessons learned and outline future directions for consumer-grade UAV-based precision agriculture applications.

**Index Terms**—consumer UAV, precision agriculture, inference servers, mobile applications, fruit detection

## I. INTRODUCTION

The role of agriculture in a country's economy is paramount, and agricultural sustainability requires better practices. Precision agriculture is a modern farming practice that aims to optimize crop production and improve yield quality using data-driven techniques [1]. One of the critical aspects of precision agriculture is to acquire accurate and timely information about crop status, such as expected yield, health, and growth [2]. This information can be beneficial for making informed decisions about irrigation, fertilization, pest control, and harvesting. Using UAVs to monitor crops offers excellent possibilities to enhance farming practices by cost-effectively acquiring field data compared to previous methods [3]. In this regard, affordable consumer-grade UAVs have great potential

to acquire field data as they can fly over large land areas quickly and capture high-resolution images of crops [4]. These images can then be processed and analyzed to extract useful crop information, such as tree count, canopy area, biomass, chlorophyll content, water stress, disease/pest infestation, and fruit yield estimation. Deep learning methods have become significantly robust in handling tasks like classification [5], detection, segmentation [6], and recommendation, making them the obvious choice for processing agricultural imagery. These techniques have been widely applied for various applications such as disease detection [7]–[9], leaf stress detection, weed detection [10], pesticide/herbicide recommendation, and yield estimation [11], [12].

However, deploying consumer-grade UAVs in precision agriculture faces challenges, such as limited battery life and a lack of onboard processing support for compute-intensive tasks. Off-the-shelf UAV-based precision agriculture solutions like DJI SmartFarm Web [13] rely on expensive and specialized hardware and software components, such as high-end cameras, GPS modules, ground control stations, and cloud servers. These components increase the cost and complexity of the UAV system and limit its accessibility and scalability for small-scale farmers.

This paper presents an efficient method to enable smartphones to serve as portable computing platforms, allowing consumer-grade UAVs to estimate yield and monitor fruits. We evaluate our pipeline on datasets of apple and peach trees collected from different orchards. We also discuss the lessons learned and the future directions for using smartphones for UAV-based precision agriculture applications.

## II. RELATED WORK

Images and other data captured by UAVs enable remote monitoring, disease/pest infestation detection [14], and yield estimation [15]. It can help farmers to plan harvesting, marketing, and storage strategies. Hyperspectral image processing has been employed to estimate yield from UAV-captured images. For instance, Wittstruck et al. [11] estimated Hokkaido

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pumpkin (*Cucurbita max.*) yield using image processing and linear regression, achieving average detection accuracy of 95%, and a Pearson correlation coefficient of 0.84 for weight estimation. Similarly, Li et al. [16] estimated potato yield using UAV-based RGB and hyperspectral images acquired at low altitudes. They used image processing to segment the green canopy regions. A random forest regression model was then used to estimate the above-ground biomass of the potato plants based on several hyperspectral features. They achieved a 0.81 correlation coefficient for yield estimation.

Most recent works use object detectors to detect visible fruits and extrapolation techniques to estimate actual fruit count. These approaches require accurate detection and counting of fruits in aerial images, which can be affected by illumination variation, occlusion, and external environmental factors. Compute-intensive deep learning-based methods are often employed to address these challenges. Zheng et al. [17] proposed an efficient online citrus counting system for large-scale unstructured orchards based on the UAV. They trained a Citrus-YOLO model to detect and count the citrus fruits in the RGB images, followed by the DeepSort algorithm to assign specific IDs to citrus fruits. Finally, they used a non-uniform distributed counter to correct the fruit count during the tracking process to reduce the counting errors caused by tracking failure. Their method achieved an F1 score of 89.07% and MAPE of 12.74%. Besides individual fruits and crops, tree health assessment has also been explored. For instance, Jemaa et al. [18] proposed a UAV-based computer vision system for orchard apple tree detection and health assessment. They used a hard negative mining strategy to detect trees. Tree health is then assessed using features extracted from 12 vegetation indices. Their system reported an F1 score of 86.24% for tree detection and an average classification accuracy of 97.52% for health assessment. Shah et al. [19] developed an Android application for drone surveillance and advanced agriculture monitoring using an EfficientNetB3 model [20]. They used a drone to capture RGB images of different crops at different growth stages. They also developed an Android application to process and analyze images using an optimized model on the smartphone device. However, they used a phone camera instead of a drone to capture the image. Egi et al. [21] proposed a drone-computer communication-based tomato generative organ counting model using YOLOv5 with the deepsort tracker. UAV captured RGB images of tomato plants at different flowering and fruiting stages within the greenhouse. Detection, tracking, and counting were performed on a PC where YOLOv5 and deepsort were used to analyze drone images. They evaluated their model on a dataset collected from a greenhouse and achieved an average F1 score of 0.63.

UAV-based precision agriculture is a promising field that can provide various benefits for crop management. However, the outstanding challenges related to robustness and efficiency require novel solutions.

### III. PROPOSED METHOD

Smartphones have become incredibly powerful in terms of computing capabilities. Their portability and inbuilt communication abilities make them an ideal platform along with UAVs for precision agriculture applications, particularly in developing countries. The proposed framework Figure 1 presents the architecture of a mobile application to analyze drone video feeds and extract useful field-related data. Most consumer-grade drones can stream camera feed via a communication channel to a smartphone, sometimes connected to a designated remote controller Table I. The camera stream is displayed on the smartphone screen in real-time as it takes flight. The video stream can be recorded on the device and later processed on the smartphone for gathering data offline. For real-time data collection, the video feed can also be received by a mobile application utilizing the UAV manufacturers' software development kit (SDK). Once the image or video feed is received, the processing pipeline implemented on-device can be invoked. However, video data processing through deep learning models is very compute-intensive, and highly efficient pipelines are needed for the smartphone application to work effectively. We propose a quantized version of a modified YOLO-Nano, a lightweight and fast object detection model with optimized hyperparameters to balance performance and efficiency. The specifics of each component are provided in the subsequent sections.

#### A. Preprocessing

Preprocessing is a crucial component in any vision-based system. In our framework, it is responsible for intensity normalization before the inference. Fruits appear very small in drone images, making it challenging for object detectors to detect them at varying scales. To cope with this problem, we augmented the labeled images by extracting smaller patches from the input image such that the bounding box annotations are relatively adjusted. This allows us to introduce many variations in object scale and orientation, essential for obtaining a scale-aware and robust detector.

#### B. Fruit Detection

Fruit detection using a deep learning model is our framework's most computationally expensive task. Though the chosen model is quite efficient, running on a video stream requires considerable computation power. To train a robust detection model that can compensate for the scale variations introduced by a drone's camera, we developed a data augmentation pipeline to introduce possible detection scenarios for the field operation of the UAV. To achieve this, overlapping patches were extracted from the labeled images using bounding box-aware transformations. The patches were again augmented using rotation, horizontal and vertical flips, and motion blur.

Object detection can be performed using a variety of efficient architectures such as Faster RCNN [26], Single shot detector (SSD) [27], You Only Look Once (YOLO) [28] and their variants. However, a lightweight single-stage detector,

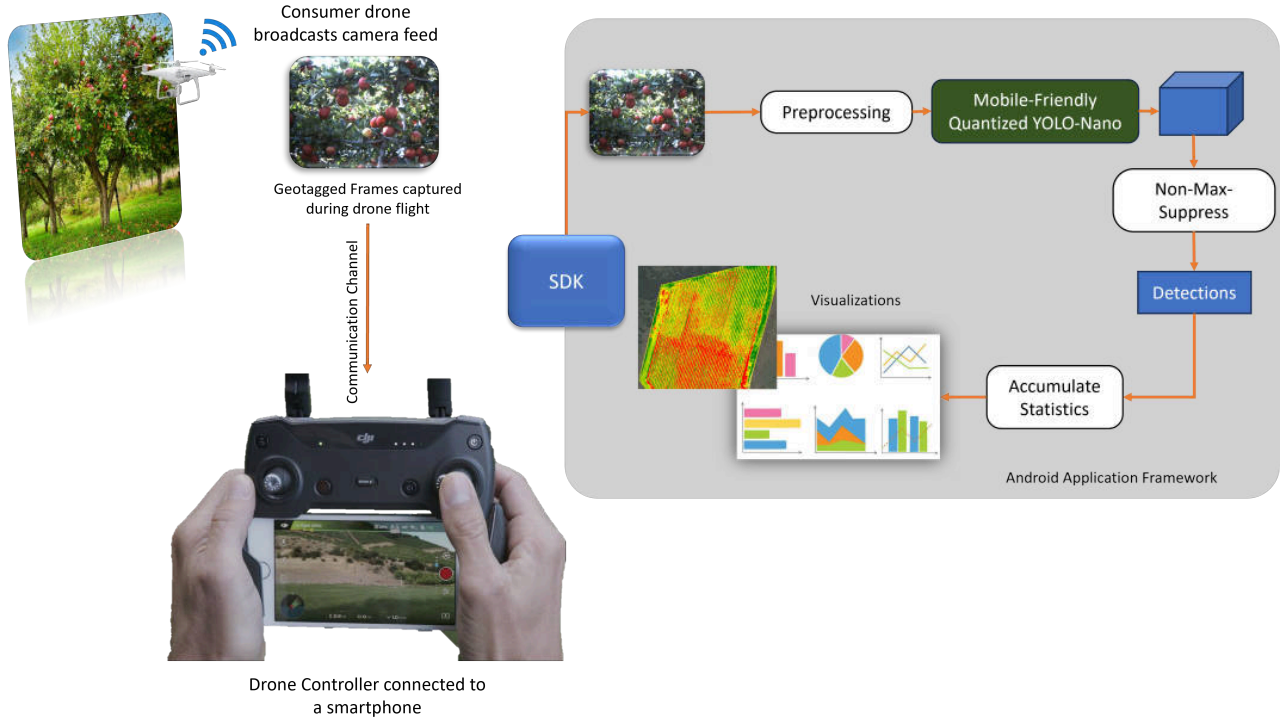


Fig. 1. Mobile application architecture for analyzing UAV images using on-board processing

TABLE I  
CONSUMER-GRADE DRONES WITH VIDEO STREAMING TO PHONE CAPABILITIES

| Model                 | Released | Price (USD) | Camera           | Stabilization | GPS | Video Range |
|-----------------------|----------|-------------|------------------|---------------|-----|-------------|
| DJI Mini 4 Pro [22]   | 2023     | \$759       | 4K/30fps, 12MP   | ✓             | ✓   | 10 km       |
| DJI Mini 3 Pro [22]   | 2022     | \$469       | 4K/30fps, 12MP   | ✓             | ✓   | 6 km        |
| Autel Evo Nano+ [23]  | 2022     | \$679       | 4K/30fps, 12MP   | ✓             | ✓   | 9 km        |
| DJI Air 2S [22]       | 2021     | \$949       | 5.4K/30fps, 20MP | ✓             | ✓   | 12 km       |
| DJI Mini 2 SE [22]    | 2020     | \$339       | 4K/30fps, 12MP   | ✓             | ✓   | 10 km       |
| Ryze Tello [24]       | 2018     | \$99        | HD/30fps, 5MP    | EIS           | ✗   | 100 m       |
| Hubsan Zino Pro+ [25] | 2019     | \$499       | 4K/30fps, 8MP    | ✓             | ✓   | 8 km        |
| Holy Stone HS175D     | 2020     | \$159       | QHD/30fps        | ✓             | ✓   | -           |
| Potensic T25 GPS      | 2018     | \$250       | FHD/25fps        | ✓             | ✓   | 300 m       |

YOLO, has been widely adopted for deployment on resource-limited hardware [29]. The YOLO version 8 family has several different architectures ranging from the most efficient YOLOv8 nano to the most potent YOLOv8x (extra large) Table II. The YOLOv8 family of models uses a backbone similar to YOLOv5. However, the cross-stage partial bottleneck with two convolutions (C2f module) is introduced instead of the CSPLayer [30]. It combines high-level features and contextual information to perform accurate detection. It also uses an anchor-free model with separate heads for detection and classification tasks. In this way, every branch focuses on its specific task, resulting in a highly optimized feature extraction pipeline. The use of complete intersection over union (CIoU) (1) along with distribution focal loss (DFL) (2) enables these models to yield superior performance with the detection of smaller objects (such as fruits). The two metrics are defined

by

$$L_{CIoU} = 1 - IoU + \frac{d^2}{C^2} + \alpha v \quad (1)$$

$$L_{DFL} = - \sum_{i=1}^N \sum_{j=1}^{N_c} y_{ij} (1 - p_{ij})^\gamma \log(p_{ij}) \quad (2)$$

where IoU is the intersection over union of the predicted and ground truth bounding boxes,  $d$  is the Euclidean distance between the center points of the boxes,  $C$  is the diagonal length of the smallest enclosing box covering the two boxes,  $v$  is a penalty term for aspect ratio consistency, and  $\alpha$  is a trade-off parameter. This loss function considers important geometrical factors relevant to the fruit detection use case, such as overlap area, normalized central point distance, and aspect ratio.  $N$  is the number of samples,  $N_c$  is the number of classes,  $y_{ij}$  is the target probability of class  $j$  for sample  $i$ , and  $p_{ij}$  is the predicted probability of class  $j$  for sample  $i$ . The parameter

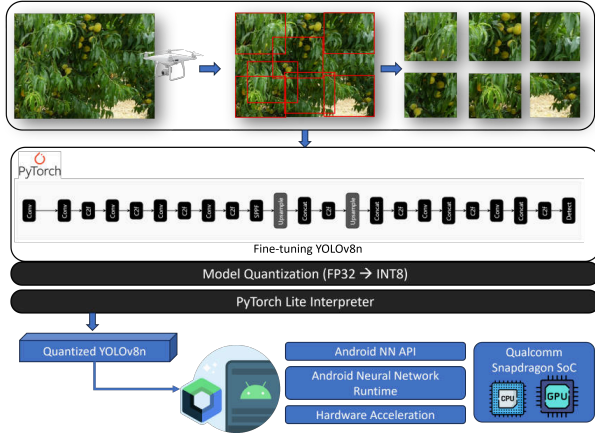


Fig. 2. Model training and optimization for deployment on smartphone device

TABLE II  
YOLOV8 VARIANTS WITH THEIR COMPUTATION LOAD

| Model         | Input (Pixels) | Params (M) | FLOPS |
|---------------|----------------|------------|-------|
| YOLOv8-Nano   | 640            | 3.2        | 8.7   |
| YOLOv8-Small  | 640            | 11.2       | 28.6  |
| YOLOv8-Medium | 640            | 25.9       | 78.9  |
| YOLOv8-Large  | 640            | 43.7       | 165.2 |
| YOLOv8-Nano-M | 640            | 2.8        | 7.6   |

$\gamma$  controls the degree of focus on hard examples. The loss function of  $L_{DFL}$  aims to align the predicted probabilities with the target probabilities, which reflects the actual distribution of the classes in the dataset. These losses enable YOLOv8 models to achieve superior detection performance at various scales.

The YOLOv8n variant consists of 7 Convolution layers along with 8 C2f modules and a spatial pyramid pooling fast (SPPF) module. We modified the width and depth of the model to obtain a lighter version YOLOv8-Nano-M and then applied post-training quantization to obtain an 8-bit fixed-point representation. It reduces the precision of the weights and activations of YOLOv8n from FP32 to INT8 to benefit from faster integer calculations of ARM SoC. It also reduces memory and power consumption by four times without sacrificing accuracy.

After quantization, the model was converted to the PyTorch Lite format, which allows it to run on mobile devices. The quantized model was then deployed on the Android application. Android platforms offer several features to speed up the inference of deep learning models on Qualcomm Snapdragon and other ARM-based SoCs. The Android OS provides a neural network application programming interface (API) to the Android neural network runtime, which provides accelerated inference on consumer hardware. The hardware acceleration is achieved using advanced features of the mobile CPU and GPU along with a dedicated digital signal processor (DSP) or a neural processing unit (NPU), commonly found in recent flagship smartphone devices Figure 2.

### C. Yield Estimation

Fruit detection is usually followed by estimating the total number of fruits on a tree, including those not visible from one side [31]. This estimation often considers additional information like canopy size, fruit density, tree height, and variety [12]. Considering the limited computing resources, we estimated the total number of fruits by summing detected fruits on individual trees and the average fruit density, and then, based on these statistics, we derived a formula to extrapolate the total fruit count  $N_F$  using

$$N_F = \sum_{n=1}^{N_t} F_d^n C_f \quad (3)$$

where  $N_t$  is the total number of trees,  $F_d^n$  is the number of detected fruits in the  $i^{th}$  tree and  $C_f$  is the correction factor to compensate for the invisible fruits or the ones that were not detected by the detector. This factor can be set adaptively based on tree height, canopy size, and average fruit density.

### D. Prototype Application Development

The prototype application for Android smartphones was developed using the pytorch-android-lite and pytorch-android-torchvision-lite libraries, enabling loading and inference on mobile devices. The overall architecture of the application is provided in Figure 1 and was built using Java for Android devices. The optimized model was transferred to a smartphone as an asset and loaded into the memory. The drone feed from its camera is streamed to the remote control connected to the smartphone. Selected frames obtained from the drone were then input to the processing pipeline Figure 3. Firstly, the input image is scaled down to the required resolution, normalized, and converted into a tensor. The model then performs inference on the input tensor and produces the output tensor containing class probabilities and bounding boxes. The post-processing module consisting of non-maximum suppression (NMS) and bounding box rescaling is applied to the output tensor to generate bounding box coordinates for the input image, which can then be overlayed on the input image for visualization. The NMS, and rescaling functions were implemented in Java. The most recently acquired frames are processed, and fruit count is collected for each tree. Some frames are dropped in the process, which is expected as video data is often highly redundant, and dropping frames won't cause significant information loss. [The app code, data, and models will be made available at: <https://github.com/jamilahmadicp/UAV-Companion-App>]

## IV. EXPERIMENTS AND RESULTS

### A. Experimental Setup

The models were trained using PyTorch deep learning framework in Python on a 10-core Xeon Server equipped with 128 GB RAM and Nvidia RTX A6000 GPU with 48 GB VRAM. The prototype Android application was installed and tested on two different Android devices: POXO X3 NFC, a mid-range device with a Snapdragon 732G CPU and Adreno 618 GPU, and Xiaomi Pad 6 with a high-end Snapdragon 870

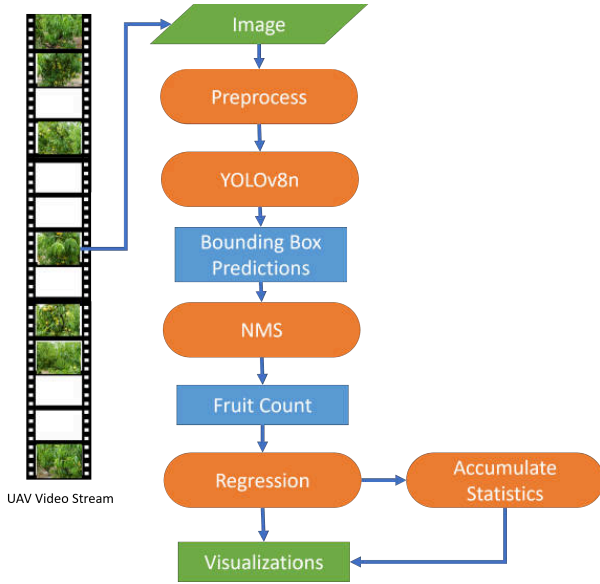


Fig. 3. Processing pipeline of the Android application for yield estimation

TABLE III  
PERFORMANCE COMPARISON OF VARIOUS MODELS ON FRUIT DETECTION

| Model         | Precision | Recall | mAP <sub>50</sub> |
|---------------|-----------|--------|-------------------|
| YOLOv8-Nano   | 0.751     | 0.701  | 0.752             |
| YOLOv8-Small  | 0.748     | 0.659  | 0.753             |
| YOLOv8-Medium | 0.738     | 0.718  | 0.751             |
| YOLOv8-Large  | 0.757     | 0.717  | 0.775             |
| YOLOv8-Nano-M | 0.752     | 0.699  | 0.750             |

CPU and Adreno 640 GPU. The performance metrics used to evaluate performance are precision, recall, and mAP<sub>50</sub>, commonly used to evaluate object detection models. Precision measures the fraction of correctly detected fruits among all detected fruits, recall measures the fraction of correctly detected fruits among all ground truth fruits, and mAP<sub>50</sub> measures the mean average precision at an intersection over union (IoU) threshold of 0.5, which is a combined measure of precision and recall across different classes of fruits. The Peach dataset consisted of 2,000 images, whereas the Apple dataset contained 1,000 images split into an 80/20 ratio for training and testing.

### B. Fruit Detection and Yield Estimation

The results presented in Table III show the performance of four different variants of the YOLOv8 family and the modified one. The variants differ in size and architectural complexity, with Nano being the smallest and fastest and Large (along with extra large) being the most potent but the slowest. YOLOv8-Large achieved the highest mAP<sub>50</sub> of 0.775 due to its more complex and capable architecture, with a larger receptive field and a higher resolution output. In contrast, YOLOv8-Nano achieved an mAP score of 0.752 with its smaller receptive field and a lower resolution output. The proposed YOLOv8-Nano-M achieved very competitive performance at a fraction of the compute cost.

With the proposed model and yield estimation method, we achieved a Pearson correlation coefficient of 0.89 for Peach fruits and 0.83 for Apples in the test datasets. These results show the approach's effectiveness in UAV-based yield estimation in stone fruits. Further research is needed to develop affordable solutions for precision agriculture involving consumer-grade UAVs, multi-sensory inputs, and smartphones as portable computing platforms.

### C. Efficiency Analysis

Evaluation results reported in Table IV show the inference time (latency) and frames per second (FPS) of different variants of the YOLOv8 family, along with the proposed quantized version of the YOLOv8-Nano model, on two different smartphone devices: SD-732G and SD-870. Inference time and the FPS depend on the size and complexity of the model, as well as the hardware specifications of the device. The results are obtained by computing the average after running the models on the test sets.

TABLE IV  
EFFICIENCY ANALYSIS OF VARIOUS MODELS ON MOBILE DEVICES

| Model             | Inference Time (s) [FPS] |             |
|-------------------|--------------------------|-------------|
|                   | SD-732G                  | SD-870      |
| YOLOv8-Nano       | 0.43 [2.3]               | 0.11 [10.2] |
| YOLOv8-Small      | 0.80 [1.3]               | 0.24 [4.2]  |
| YOLOv8-Medium     | 1.10 [0.9]               | 0.63 [1.6]  |
| YOLOv8-Large      | 1.44 [0.7]               | 0.74 [1.4]  |
| YOLOv8-Nano-416   | 0.23 [4.3]               | 0.07 [14.3] |
| YOLOv8-Nano-320   | 0.14 [7.2]               | 0.06 [16.6] |
| YOLOv8-Nano-M-320 | 0.11 [9.1]               | 0.05 [20.0] |

The non-quantized variants of the models performed as expected. The YOLOv8-Nano, being the fastest, was further optimized through quantization and input resolution reduction, which caused it to achieve nearly 10 FPS on the mid-range device compared to the non-quantized 2.3 FPS. Similarly, its performance on SD-870 also improved from 10 FPS to 20+ FPS after quantization and input resolution reduction, making it suitable for real-time inference on UAV video streams.

The other more capable models can also be quantized and optimized for more recent and capable hardware to achieve better performance.

## V. DISCUSSION

Enabling consumer-grade UAVs for precision agriculture applications through mobile computing platforms offers many advantages. They are inexpensive compared to specialized UAVs for precision agriculture, which have dedicated computing hardware onboard. Drone footage can be stored on the smartphone device for future analysis, allowing for assessments of the effectiveness of fertilizers or pesticides at a later time. Similarly, yield estimate data can be aggregated from videos captured at different dates during a season to obtain more accurate statistics. Insights extracted from the UAV video stream can be viewed on the smartphone or uploaded to the cloud for future use. In brief, utilizing consumer-grade UAVs with smartphone applications capable of analyzing video feeds

will enable many farmers to adopt innovative methods that may not have the resources for expensive off-the-shelf solutions.

## VI. CONCLUSION

This paper presented an efficient processing pipeline to analyze UAV-captured video streams of orchards for yield estimation. We demonstrated how video streams from UAVs equipped with RGB cameras can be processed on smartphones or edge devices using preprocessing, object detection, and yield estimation. Our evaluation of apple and peach datasets illustrated their effectiveness in detecting and counting fruits and estimating yield. Our method can potentially reduce the cost and complexity of deploying UAVs for precision agriculture, particularly in small-to-medium-sized fields and developing countries. Additionally, we discussed the challenges and opportunities for future research in this domain, including improving the accuracy and robustness of the fruit detector, integrating other sensors and data sources, and creating user-friendly interfaces and applications.

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## REFERENCES

- [1] S. Coulibaly, B. Kamsu-Foguem, D. Kamissoko, and D. Traore, "Deep learning for precision agriculture: A bibliometric analysis," *Intelligent Systems with Applications*, vol. 16, p. 200102, 2022.
- [2] A. Dakir, F. Barramou, and O. B. Alami, "Opportunities for artificial intelligence in precision agriculture using satellite remote sensing," *Geospatial Intelligence: Applications and Future Trends*, pp. 107–117, 2022.
- [3] M. F. Aslan, A. Durdu, K. Sabanci, E. Ropelewska, and S. S. Gültekin, "A comprehensive survey of the recent studies with uav for precision agriculture in open fields and greenhouses," *Applied Sciences*, vol. 12, no. 3, p. 1047, 2022.
- [4] N. Lu, J. Zhou, Z. Han, D. Li, Q. Cao, X. Yao, Y. Tian, Y. Zhu, W. Cao, and T. Cheng, "Improved estimation of aboveground biomass in wheat from rgb imagery and point cloud data acquired with a low-cost unmanned aerial vehicle system," *Plant Methods*, vol. 15, no. 1, pp. 1–16, 2019.
- [5] Z. Dai, H. Liu, Q. V. Le, and M. Tan, "Coatnet: Marrying convolution and attention for all data sizes," *Advances in neural information processing systems*, vol. 34, pp. 3965–3977, 2021.
- [6] S. Minaee, Y. Boykov, F. Porikli, A. Plaza, N. Kehtarnavaz, and D. Terzopoulos, "Image segmentation using deep learning: A survey," *IEEE transactions on pattern analysis and machine intelligence*, vol. 44, no. 7, pp. 3523–3542, 2021.
- [7] J. Ahmad, B. Jan, H. Farman, W. Ahmad, and A. Ullah, "Disease detection in plum using convolutional neural network under true field conditions," *Sensors*, vol. 20, no. 19, p. 5569, 2020.
- [8] F. A. Shah, M. A. Khan, M. Sharif, U. Tariq, A. Khan, S. Kadry, and O. Thinnukool, "A cascaded design of best features selection for fruit diseases recognition," *Comput. Mater. Contin.*, vol. 70, pp. 1491–1507, 2022.
- [9] M. Zia Ur Rehman, F. Ahmed, M. Attique Khan, U. Tariq, S. Shaukat Jamal, J. Ahmad, and I. Hussain, "Classification of citrus plant diseases using deep transfer learning," *Computers, Materials & Continua*, vol. 70, no. 1, 2021.
- [10] J. Ahmad, K. Muhammad, I. Ahmad, W. Ahmad, M. L. Smith, L. N. Smith, D. K. Jain, H. Wang, and I. Mehmood, "Visual features based boosted classification of weeds for real-time selective herbicide sprayer systems," *Computers in Industry*, vol. 98, pp. 23–33, 2018.
- [11] L. Wittstruck, M. Schirrmann, U. Lussem, J. B"urger, and A. Ruckelshausen, "Uav-based rgb imagery for hokkaido pumpkin (cucurbita max.) detection and yield estimation," *Sensors*, vol. 21, no. 1, p. 118, 2020.
- [12] M. Mokria, A. Gebrekirstos, H. Said, K. Hadgu, N. Hagazi, W. Dubale, and A. Bräuning, "Fruit weight and yield estimation models for five avocado cultivars in ethiopia," *Environmental Research Communications*, vol. 4, no. 7, p. 075013, 2022.
- [13] D. Agriculture, "DJI smartfarm web," 2023. [Online]. Available: <https://ag.dji.com/smartfarm-web>
- [14] A. Albanese, M. Nardello, and D. Brunelli, "Automated pest detection with dnn on the edge for precision agriculture," *IEEE Journal on Emerging and Selected Topics in Circuits and Systems*, vol. 11, no. 3, pp. 458–467, 2021.
- [15] T. B. Shahi, C.-Y. Xu, A. Neupane, and W. Guo, "Recent advances in crop disease detection using uav and deep learning techniques," *Remote Sensing*, vol. 15, no. 9, p. 2450, 2023.
- [16] B. Li, X. Zhang, Z. Li, J. Liu, L. Liu, and G. Yang, "Above-ground biomass estimation and yield prediction in potato by using uav-based rgb and hyperspectral imaging," *ISPRS Journal of Photogrammetry and Remote Sensing*, vol. 162, p. 161–172, 2020.
- [17] Z. Zheng, Y. Wang, S. Wang, Z. Zhang, H. Zhang, X. Chen, and Y. Zhang, "An efficient online citrus counting system for large-scale unstructured orchards based on the unmanned aerial vehicle," *Journal of Field Robotics*, vol. 40, no. 3, p. 552–573, 2023.
- [18] H. Jemaa, M. A. Ben Abdallah, B. Bouaziz, and M. E. M. Mountassir, "Uav-based computer vision system for orchard apple tree detection and health assessment," *Remote Sensing*, vol. 15, no. 14, p. 3558, 2023.
- [19] S. A. Shah, G. M. Lakho, H. A. Keerio, M. N. Sattar, G. Hussain, M. Mehdi, R. B. Vistro, E. A. Mahmoud, and H. O. Elansary, "Application of drone surveillance for advance agriculture monitoring by android application using convolution neural network," *Agronomy*, vol. 13, no. 7, p. 1764, 2023.
- [20] M. Tan and Q. Le, "Efficientnet: Rethinking model scaling for convolutional neural networks," in *International conference on machine learning*. PMLR, 2019, pp. 6105–6114.
- [21] Y. Egi, M. Hajyzadeh, and E. Eyceyurt, "Drone-computer communication based tomato generative organ counting model using yolo v5 and deep-sort," *Agriculture*, vol. 12, no. 9, p. 1290, 2022.
- [22] DJI, "DJI mobile sdk - downloads," <https://developer.dji.com/mobile-sdk/downloads/>, 2021, accessed on 2023-10-06.
- [23] A. Robotics, "Autelsdk - github," <https://github.com/AutelSDK>, 2021, accessed on 2023-10-06.
- [24] R. Tech, "Tello - downloads," <https://www.ryzerobotics.com/tello/downloads>, 2021, accessed on 2023-10-06.
- [25] Hubsan, "Connect tutorial - hubsan zino series," <https://www.hubsan.com/support/files/products/ZINO/ct/connectTutorial.pdf>, 2019, accessed on 2023-10-08.
- [26] R. Girshick, "Fast r-cnn," in *Proceedings of the IEEE international conference on computer vision*, 2015, pp. 1440–1448.
- [27] W. Liu, D. Anguelov, D. Erhan, C. Szegedy, S. Reed, C.-Y. Fu, and A. C. Berg, "Ssd: Single shot multibox detector," in *Computer Vision—ECCV 2016: 14th European Conference, Amsterdam, The Netherlands, October 11–14, 2016, Proceedings, Part I 14*. Springer, 2016, pp. 21–37.
- [28] J. Terven and D. Cordova-Esparza, "A comprehensive review of yolo: From yolov1 to yolov8 and beyond," *arXiv preprint arXiv:2304.00501*, 2023.
- [29] X. Zheng and T. He, "Reduced-parameter yolo-like object detector oriented to resource-constrained platform," *Sensors*, vol. 23, no. 7, p. 3510, 2023.
- [30] J. Terven and D. Cordova-Esparza, "A comprehensive review of yolo: From yolov1 and beyond," 2023.
- [31] F. Gao, W. Fang, X. Sun, Z. Wu, G. Zhao, G. Li, R. Li, L. Fu, and Q. Zhang, "A novel apple fruit detection and counting methodology based on deep learning and trunk tracking in modern orchard," *Computers and Electronics in Agriculture*, vol. 197, p. 107000, 2022.